

MATH 114 HW 2

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Question 1

Define the sample mean $\bar{X}_n$ and sample variance S_n^2 as

$$\bar{X}_n = \frac{1}{n} \sum_{k=1}^n X_k \quad S_n^2 = \frac{1}{n-1} \sum_{k=1}^n (X_k - \bar{X}_n)^2$$

Recall $\sigma^2 = \mathbb{E}(X^2) - \mu^2 \Rightarrow \sigma^2 - \mu^2 = \mathbb{E}(X^2)$

$$\begin{aligned} S_n^2 &= \frac{1}{n-1} \sum_{k=1}^n (X_k - \bar{X}_n)^2 \\ &= \frac{1}{n-1} \left(\sum_{k=1}^n X_k^2 - n\bar{X}_n^2 \right) \\ \mathbb{E}(S_n^2) &= \frac{1}{n-1} \left(\sum_{k=1}^n \mathbb{E}(X_k^2) - n\mathbb{E}(\bar{X}_n^2) \right) \\ &= \frac{1}{n-1} \left(\sum_{k=1}^n (\mu^2 - \sigma^2) - n((\mathbb{E}(\bar{X}))^2 + \text{Var}\bar{X}) \right) \\ &= \frac{1}{n-1} \left(n\mu^2 + n\sigma^2 - n\left(\mathbb{E}\left(\frac{1}{n} \sum_{k=1}^n X_k\right)^2 + \text{Var}\left(\frac{1}{n} \sum_{k=1}^n X_k\right)\right) \right) \\ &= \frac{1}{n-1} \left(n\mu^2 + n\sigma^2 - n\left(\left(\frac{1}{n} \sum_{k=1}^n \mathbb{E}(X_k)\right)^2 + \left(\frac{1}{n^2} \sum_{k=1}^n \text{Var}(X_k)\right)\right) \right) \\ &= \frac{1}{n-1} \left(n\mu^2 + n\sigma^2 - n\left(\left(\frac{1}{n} \cdot n\mu\right)^2 + \frac{1}{n^2} \cdot n\sigma^2\right) \right) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{n-1} (n\mu^2 + n\sigma^2 - n\mu^2 - \sigma^2) \\
&= \frac{1}{n-1} (n-1)(\sigma^2) \\
&= \sigma^2
\end{aligned}$$

Question 2

Define $I = \mathbb{E}(g(X))$ and

$$I = \int_D g(x)f(x)dx \qquad I_n = \frac{1}{n} \sum_{k=1}^n g(X_k)$$

We have

$$\begin{aligned}
\text{Var}(I_n) &= \text{Var}\left(\frac{1}{n} \sum_{k=1}^n g(X_k)\right) \\
&= \frac{1}{n^2} \sum_{k=1}^n \text{Var}(g(X_k)) \\
&= \frac{1}{n} \text{Var}(g(X)) \\
&= \frac{1}{n} (\mathbb{E}(g(X)^2) - \mathbb{E}(g(X))^2) \\
&= \frac{1}{n} \left(\int_D g(x)^2 f(x) dx - I^2 \right)
\end{aligned}$$

Question 3

N	I_N	$\text{Var}(I_N)$	J_N	$\text{Var}(J_N)$
100	0.8107	0.0026	0.8276	0.0017
400	0.8177	0.00086	0.8527	0.00036
900	0.8423	0.00029	0.8266	0.00020
1600	0.8174	0.00017	0.8397	0.000079

Code for the simulation is included below.

```
import numpy as np
N = 100
def crude_monte_carlo_integration():
```

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X_n = np.random.uniform(0, 4, N)
sum = 0
for x in X_n:
    sum += 4 * (np.sqrt(x) * np.exp(-x))
return 1 / N * sum

def importance_sampling_integration():
    U = np.random.uniform(0, 1, N)
    Y_n = -np.log(1 - U * (1 - np.exp(-4)))

    sum = 0
    for y in Y_n:
        sum += (np.sqrt(y) * np.exp(-y)) / (np.exp(-y) / (1 - np.exp(-4)))
    return 1 / N * sum

def samp_var(X):
    sum = 0
    for i in range(len(X)):
        sum += (X[i] - np.mean(X)) ** 2
    return sum / (len(X) - 1)

results_mc = []
for i in range(1, 101):
    results_mc.append(crude_monte_carlo_integration())
var_mc = samp_var(results_mc)

results_is = []
for i in range(1, 101):
    results_is.append(importance_sampling_integration())
var_is = samp_var(results_is)
crude_monte_carlo_integration(), var_mc, importance_sampling_integration(), var_is

```

Question 4

a. Define $g(x) = \frac{1}{1+x^2}$ and $I = \int_0^1 g(x)dx$

$$\begin{aligned}
 I_N &= \frac{1}{2N} \sum_{k=1}^n (g(X_k) + g(Y_k)) \\
 &= \frac{1}{2N} \left(\sum_{k=1}^n g(X_k) + \sum_{k=1}^n g(Y_k) \right) \\
 \mathbb{E}(I_N) &= \frac{1}{2N} \left(\sum_{k=1}^n \mathbb{E}(g(X_k)) + \sum_{k=1}^n \mathbb{E}(g(Y_k)) \right) \\
 &= \frac{1}{2N} \left(\sum_{k=1}^n I + \sum_{k=1}^n I \right) \\
 &= \frac{1}{2N} (NI + NI) \\
 &= I
 \end{aligned}$$

$$\begin{aligned}
 A_N &= \frac{1}{2N} \sum_{k=1}^n (g(X_k) + g(1 - X_k)) \\
 &= \frac{1}{2N} \left(\sum_{k=1}^n g(X_k) + \sum_{k=1}^n g(1 - X_k) \right) \\
 \mathbb{E}(A_N) &= \frac{1}{2N} \left(\sum_{k=1}^n \mathbb{E}(g(X_k)) + \sum_{k=1}^n \mathbb{E}(g(1 - X_k)) \right) \\
 &= \frac{1}{2N} \left(\sum_{k=1}^n I + \sum_{k=1}^n \int_0^1 g(1 - X) \cdot 1 dx \right) \\
 &= \frac{1}{2N} \left(NI + \sum_{k=1}^n \int_0^1 \frac{1}{1 + (1 - X)^2} dx \right)
 \end{aligned}$$

$$\begin{aligned}
 U &= 1 - X \\
 dU &= -dX
 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2N} \left(NI + \sum_{k=1}^n \int_0^1 \frac{1}{1+(U)^2} du \right) \\
&= \frac{1}{2N} \left(NI + \sum_{k=1}^n I \right) \\
&= \frac{1}{2N} (NI + NI) \\
&= I
\end{aligned}$$

b. skipped

c.

N	I_N	$\text{Var}(I_N)$	J_N	$\text{Var}(J_N)$
100	3.1627	0.0015	3.1406	0.000036
400	3.1818	0.00040	3.1396	0.0000067
900	3.1522	0.00030	3.1404	0.0000030
1600	3.1163	0.00011	3.1415	0.0000019

Below is the code for this simulation.

```

import numpy as np
N = 1600
def crude_monte_carlo_integration2():
    X_n = np.random.uniform(0, 1, N)
    Y_n = np.random.uniform(0, 1, N)
    sum = 0
    for i in range(N):
        sum += (1/(1+X_n[i]**2)) + (1/(1+Y_n[i]**2))
    return 2/N * sum

def antithetic_integration():
    X_n = np.random.uniform(0, 1, N)
    sum = 0
    for i in range(N):
        sum += (1/(1+X_n[i]**2)) + (1/(1+(1-X_n[i])**2))

```

```

        return 2/N * sum

results_mc2 = []
for i in range(1, 101):
    results_mc2.append(crude_monte_carlo_integration2())
var_mc2 = samp_var(results_mc2)

results_anti = []
for i in range(1, 101):
    results_anti.append(antithetic_integration())
var_anti = samp_var(results_anti)
crude_monte_carlo_integration2(), var_mc2, antithetic_integration(), var_anti

```